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# G2RINS: A Generative String-and-Graph Polymer Representation to Assist Computational Materials Discovery

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## Abstract

There is considerable interest in developing generative models for the design of new polymeric materials. A major challenge towards the development of such models is designing machine-readable representations that capture the hierarchical and stochastic nature of polymers. In this work, we introduce G2RINS, a generative line and graph notation that builds on G-BigSMILES to account for the multistep hierarchical microstructure of real polymers by properly describing prepolymers and branching. This new string representation considerably improves the descriptive capabilities of the overall framework. Additionally, our dedicated Python implementation automatically converts G2RINS strings into generative graphs to create meaningful polymer chain ensembles. Together, the G2RINS string, graph, and generated ensembles serve as versatile tools for database labeling, high-throughput molecular dynamics (MD) simulations, machine learning (ML) modeling, and fingerprinting. Ultimately, G2RINS provides a multilevel, machine-readable representation with the potential to dramati-

## Keywords

Polymer informatics, Generative Representation, Graph Representation, Chain Ensemble, Line Notation, BigSMILES, SMILES

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## Abstract

There is considerable interest in developing generative models for the design of new polymeric materials. A major challenge towards the development of such models is designing machine-readable representations that capture the hierarchical and stochastic nature of polymers. In this work, we introduce G<sup>2</sup>RINS, a generative line and graph notation that builds on G-BigSMILES to account for the multistep hierarchical microstructure of real polymers by properly describing prepolymers and branching.

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<sup>1</sup>These authors contributed equally to this work.

overall framework. Additionally, our dedicated Python implementation automatically converts G<sup>2</sup>RINS strings into generative graphs to create meaningful polymer chain ensembles. Together, the G<sup>2</sup>RINS string, graph, and generated ensembles serve as versatile tools for database labeling, high-throughput molecular dynamics (MD) simulations, machine learning (ML) modeling, and fingerprinting. Ultimately, G<sup>2</sup>RINS provides a multilevel, machine-readable representation with the potential to dramatically accelerate the discovery of new polymer materials.

## 1. Introduction

Polymers play a vital role in multiple industries, from packaging to advanced healthcare technologies, directly impacting daily life.<sup>1-3</sup> There is a growing demand for sustainable polymer materials with customized properties for specific applications. In this context, modern computational approaches, such as artificial intelligence (AI) and machine learning (ML), are gaining significant attention due to their potential to drastically accelerate polymer material discovery.<sup>4-6</sup>

Creating machine-readable polymer representations is central to the development of AI-assisted tools for the discovery of polymeric materials. Digital representations serve as identifiers in databases<sup>7,8</sup> and as input for ML models for property prediction;<sup>9</sup> they are the starting point of automated pipelines for molecular simulations for the prediction of material properties<sup>10,11</sup> and can also be used to improve query capabilities through datasets for similarity and substructure search.<sup>6,12</sup> However, a universal representation that truly comprehends the full complexity of a polymer material remains an open challenge.

Polymers are formed by numerous repeating units that are arranged in different possible architectures (e.g., linear, star, branched).<sup>13,14</sup> Synthetic pathways to create polymer materials yield a stochastic mixture of different molecules rather than a unique molecular structure. These polymer chains can vary in composition, molecular weight, sequence, and even architecture, posing difficulties in the development of concise yet complete represen-

tations. An additional challenge arises from the fact that polymers are often the result of a multi-step polymerization process, leading to complex polymer architectures formed by nested statistical distributions of polymer chains.

In recent years, several attempts to develop polymer representations have been published, including line notations (P-SMILES,<sup>15</sup> CurlySMILES,<sup>16</sup> InChi,<sup>8</sup> HELM<sup>17</sup>) and graph representations,<sup>18–21</sup> among others.<sup>7</sup> While text-based notations are a natural choice as the starting point for polymer informatics, graphs have recently gained attention due to their efficiency as inputs for ML models based on Graph Neural Networks (GNNs).<sup>22</sup> Graphs are also a convenient way to organize data for diverse polymer informatics tasks, such as the derivation of fingerprints and similarity scores.<sup>23</sup> However, none of the available representations comprehensively capture the stochastic nature and architectural complexity of polymer materials.

Traditional line notations describe unique molecular structures atom by atom, a suitable approach for single molecules but not for stochastic molecular entities. The Simplified Molecular-Input Line-Entry System (SMILES)<sup>24,25</sup> is a popular choice in computational chemistry due to its simplicity and straightforward usability. Among the several attempts to extend SMILES for polymer description, BigSMILES<sup>26</sup> represents a significant step-forward. This SMILES-based notation introduces features such as bond connectors between repeat units and a hierarchical notation to account for the stochastic nature of polymers. BigSMILES is gaining popularity among the polymer informatics community due to its simplicity inherited from SMILES and its ability to represent diverse polymer structures.<sup>9,27–29</sup> However, this notation does not consider the reactivity of connection sites or the molecular weight distribution of polymer chains, and can only qualitatively describe the nested microstructure of polymers. Not only do these properties significantly impact the performance of the material, but they are also essential for generating chain ensembles that are representative of the actual polymer. Having access to accurate polymer ensembles would enable a new level of polymer representation for polymer informatics, with the po-

tential to streamline the automated setup of molecular simulations and the development of ensemble-based embeddings for machine learning and artificial intelligence.

Recently, our group developed Generative BigSMILES (G-BigSMILES).<sup>30</sup> G-BigSMILES extends BigSMILES to incorporate ensemble information as well as the likelihood of connections between repeat units. Due to these new grammar features, the notation includes sufficient information to generate polymer molecules that are representative of the polymer ensemble. Our G-BigSMILES implementation<sup>31</sup> allows the creation of a graph that encodes the information found in a G-BigSMILES string. The graph is a generative object capable of creating statistically accurate ensembles of polymer chains and can potentially be connected to ML models and MD automatic pipelines. Even though G-BigSMILES is the first generative line notation that can represent a polymer material in a complete way, the scope of polymers that can be described is limited due to grammatical constraints. Furthermore, G-BigSMILES only considers polymers synthesized in a single step, further reducing the number of polymer architectures that can be represented.

Here, we introduce the **Generative Graph Representation of Integrated Nested BigSMILES** (G<sup>2</sup>RINS, also written GGRINS; pronounced ‘grins’), a line notation that builds upon G-BigSMILES. G<sup>2</sup>RINS introduces new grammatical features while maintaining the logic of the previous SMILES-based notations. The renewed syntax allows for the description of more complex polymer architectures using simpler grammar. G<sup>2</sup>RINS incorporates nested stochastic objects that represent polymers synthesized through multiple steps of polymerization reactions. Thus, the G<sup>2</sup>RINS string encodes the full synthetic pathway of a polymer material and is capable of generating individual polymer molecules that are statistically representative of the real polymer ensemble. In addition, we present the G<sup>2</sup>RINS graph, a generative graph that stores all the information found in a G<sup>2</sup>RINS string. The G<sup>2</sup>RINS graph is inspired by its G-BigSMILES counterpart, but it is reshaped and enhanced to account for the novel G<sup>2</sup>RINS capabilities. The graph makes G<sup>2</sup>RINS executable by generating accurate polymer ensembles. In addition, it can be directly connected to GNNs-based ma-

chine learning models. Finally, G<sup>2</sup>RINS is also a code<sup>32</sup> that performs syntactic and semantic validations on a user-provided G<sup>2</sup>RINS string and automatically handles the creation of the corresponding generative graph and the generation of polymer ensembles.

The remainder of this article is structured as follows. We first detail the grammatical features of the G<sup>2</sup>RINS notation (Section 2), followed by a description of the G<sup>2</sup>RINS graph and the chain generation process (Sections 3 and 4). To validate this framework, Section 5 illustrates its application to polymers with complex architectures and synthetic pathways. We conclude by discussing how G<sup>2</sup>RINS can accelerate the discovery of new polymer materials and outlining the future challenges for the notation (Section 6), before summarizing our key findings (Section 7).

## 2. General Description of the G<sup>2</sup>RINS Notation

### 2.1. The Stochastic Object

G<sup>2</sup>RINS grammar is built as an extension of the BigSMILES and G-BigSMILES rules. The elemental G<sup>2</sup>RINS grammar object is the "stochastic object" enclosed by curly brackets; see the general formula in Figure 1a. The stochastic object comprises three main elements, separated by semicolons: the list of repeat units, the list of initiators, and the list of terminators. This represents a novelty with respect to BigSMILES and G-BigSMILES, which consider a single list of end groups. Having terminators and initiators in separate lists significantly simplifies the description of polymers with multiple initiators and terminators with differentiated reactivity and enables the representation of multi-step polymers. A fourth element describes the ensemble general properties (e.g., the molecular weight distribution) and is defined outside the curly brackets, enclosed by vertical bars. A stochastic object represents the result of a single polymerization reaction, considering the connectivity of initiators, repeat units, and terminators, as well as the resulting molecular weight distribution. Thus, it encodes the information to create an ensemble of chains that represents the real polymer.

**(a) General Formula**

{ repeat units ; initiator ; terminator } | ensemble descriptors |

**(b) Bond Connector Syntax**

| Type           | Syntax        | With Probability    | Features              |
|----------------|---------------|---------------------|-----------------------|
| Directional    | [<][>]        | [< p ][> p ]        | Conjugated connection |
| Nondirectional | [\$]          | [\$ p ]             | Symmetric connection  |
| Indexed        | [<1][>2]      | [<1 p ][>2 p ]      | Bond ID               |
| Combined       | [<,>][\$1,<2] | [<,> p ][\$1,<2 p ] | Multiple connection   |

**(c) Example: poly(ethylene oxide-ran-propylene oxide)**

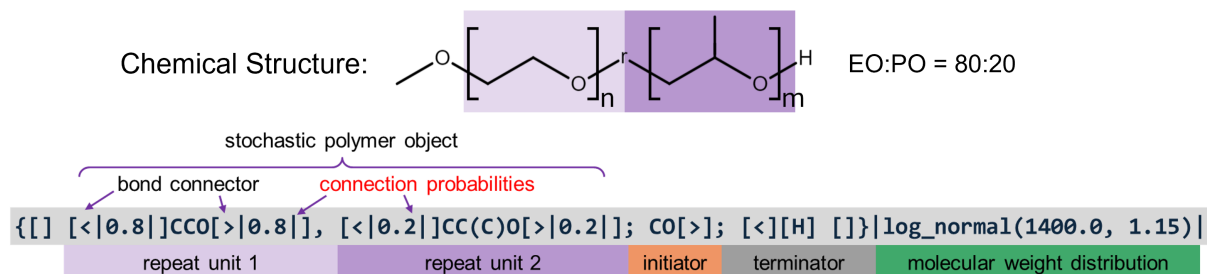


Figure 1: (a) General formula of a G<sup>2</sup>RINS string. (b) Description of the different types of bond connectors, their grammar, and their function. (c) Example depicting an ethylene-glycol:propylene-glycol 80:20 copolymer and the corresponding G<sup>2</sup>RINS string. The different elements of the string are labeled using the same color code than in the general formula. The same polymer can be described using relative molar amounts as described in section 2.3: {[[] [<]CCO[>]|.8|, [<]CC(C)O[>]|.2|; CO[>]; [<][H] []]|log\_normal(1400,1.15)|

## 2.2. Bond Connectors and Connection Probabilities

The repeat units, initiators, and terminators in a stochastic object are SMILES strings encoding their molecular structures and are connected through bond descriptors or bond connectors, as in BigSMILES and G-BigSMILES. Bond connectors are enclosed by square brackets, as shown in Figure 1b. There are two main types of bond connectors. Directional bond connectors are defined by the symbols  $<$  and  $>$ . Only conjugated bonds,  $[<]$  and  $[>]$ , will connect. Non-directional bond connectors are defined by the symbol  $\$$  and represent a symmetric connection: a  $[\$]$  bond connector will pair with any other  $[\$]$  bond connector, including itself.

Bond connectors can have ID numbers after the bond symbol. Only bond connectors with the same ID number will connect. The default ID number is 0.

The bond connectors have a connection weight or probability, enclosed by vertical bars right after the symbol list and represented by the letter  $p$  in Figure 1b. The default value is 1.0. This number reflects the reactivity of the connection site for the generation process. During generation, these connection probabilities will be adequately normalized and considered at each growth step to account for the reactivity of the possible connections.  $p$  can also be a matrix of connection probabilities as explained above in section 2.3.

To illustrate the usage of bond connectors, Figure 1c shows an example of a G<sup>2</sup>RINS string that represents a simple linear copolymer formed by units of ethylene oxide (EO) and propylene oxide (PO) in an 80:20 proportion.<sup>33</sup> The initiator is a methanol group that can connect to the C end of any repeat unit. The repeat units have directional connectors that define the growth of the chain: the O end of either the EO or PO unit connects to the C end of any of the units. The bond weights are defined to encode the proper probability of growing an EO or a PO unit. Finally, upon termination, the units will be capped with a hydrogen atom through their O end.

In G<sup>2</sup>RINS, we introduce the possibility of having combined bond connectors, where more than one symbol can be defined in the same bond connector, separated by commas, as shown

in Figure 1b. For example, [ $\$, <1$ ] will connect with any other bond connector that contains the  $\$$  and/or the  $>1$  symbol in its list of symbols. This new feature significantly expands the range of representable polymer structures, as it allows a chemical group to connect to multiple other groups that are not necessarily linked to one another. For example, in the following string representing alternating poly(lactic acid-alt-glycolic acid),<sup>34</sup> the same terminator is connected to both repeat units through the use of a list of bond symbols:

```
{[ ] [<]OC(C)C(=O)[>1], [<1]OCC(=O)[>];  
[H][>]; [<,<1]O [ ]}|poisson(5000)|
```

G<sup>2</sup>RINS allows an atom to have more than one bond connector. Although this is contemplated in BigSMILES, it is forbidden in G-BigSMILES due to implementation issues. An atom with more than one bond connector can potentially grow one chain per bond connector. This is particularly useful for describing polymers with repeat units or initiators comprising atoms with two or more reactive sites. For example, the G<sup>2</sup>RINS repeat unit for branched polyethylene imine<sup>35</sup> is

```
[<]CCN([>])[>]
```

where each N atom has two [ $>$ ] bond connectors. This reflects the ability of the atom to react twice with a compatible repeat unit.

### 2.3. Refinement of Connection Probabilities: Tunable Connection Probabilities Matrix and Molar Amounts

To further control the probability of possible connections, G<sup>2</sup>RINS provides two optional grammatical features. The first, already available in G-BigSMILES,<sup>30</sup> consists of defining the matrix of specific connection probabilities for the bond connectors. This is done by specifying in bond connectors the connection probability with any other bond connector in the stochastic object in the order of appearance. To illustrate this, Figure 2a (top) shows

a schematic string to describe block copolymers with tunable blockiness.<sup>36</sup> In the string, monomers A and B have bond connectors with defined probabilities for their connection ( $p_{AA}$  and  $p_{BB}$ ) and to the other monomer ( $p_{AB}$  and  $p_{BA}$ ). Zeros are required for non compatible connections. Thus, alternating, statistical, and block polymers can be described by tuning  $p_{AA}$ ,  $p_{BB}$ ,  $p_{AB}$ , and  $p_{BA}$ , as shown in the examples provided in Figure 2a. Gradient polymers are also accessible by including probabilities that vary with chain growth, but this possibility has not yet been included in our G<sup>2</sup>RINS python implementation.<sup>32</sup>

The second feature, new to G<sup>2</sup>RINS, allows the definition of relative molar amounts for the repeat units, initiators, and terminators. This is done by adding a number enclosed by vertical bars after each unit in the string, as shown in the following string:

```
{[] [<]CCO[>]|8|, [<]CC(C)O[>]|2|;
CO[>]; [<][H] []}|log_normal(1400, 1.15)|
```

The molar amount will affect the probability of growing a repeat unit in the generating process. In the example, picking a CCO unit over a CC(C)O has a weight of 8 versus 2. In generation, this weight is multiplied by the connection probabilities defined in the bond connectors. This allows for the separate definition of the reactivity of a site and the feed ratio of the repeat unit to which that site belongs. In a generic poly(ethylene glycol) repeat unit  $[<|p_1|]CCO[>|p_2|]|n|$ ,  $p_1$  and  $p_2$  should reflect the reactivity of the corresponding reactive sites, and  $n$  should reflect the feed ratio of the unit. The resulting non-normalized weight of adding this unit to a growing chain from the C side will be  $w = n \cdot p_1$ .

## 2.4. Ensemble Descriptors: Molecular Weight Distributions

The molecular weight distribution of a stochastic object is defined within vertical bars, outside the curly brackets enclosing the repeat units, initiators, and terminators. In principle, any description of the ensemble distribution could be placed in this part of the string, including experimental data. At the moment of the publication of this article, the implemen-

**(a) Sequence control via connection probabilities matrix**

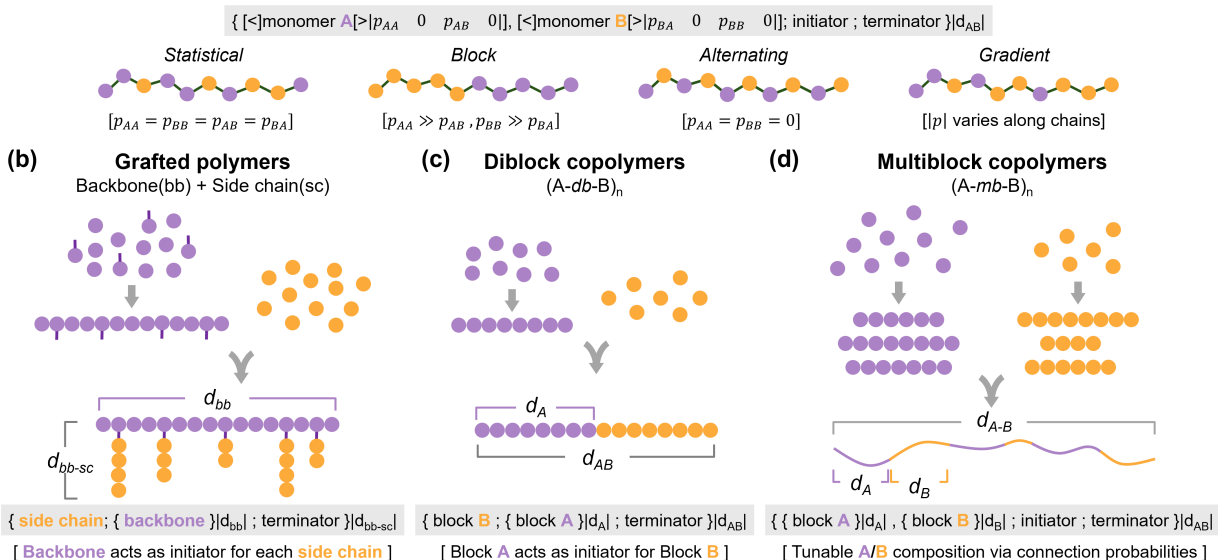


Figure 2: (a) G<sup>2</sup>RINS schematic string for block copolymer with tunable blockiness and examples of representable structures. Schemes of the synthetic pathway and the corresponding G<sup>2</sup>RINS nested structure of stochastic objects for (a) graft polymers, (b) diblock copolymers, and (c) multiblock copolymers.

tation only supports the model distributions that are already available in G-BigSMILES:<sup>30</sup> Gaussian, Uniform, Poisson, Flory-Schulz, Schulz-Zimm, and Log-normal. For example, a Log-normal distribution will be defined in a G<sup>2</sup>RINS string as  $| \log\_normal(M_n, D) |$ , where  $M_n$  represents the number-average molecular weight of the chains and  $D$  the polydispersity index. In the example shown in Figure 1, the stochastic object has a Log-normal molecular weight distribution with  $M_n = 1400$  and  $D = 1.15$ .

For more details on the implementation of the model molecular weight distributions, we refer the reader to the G-BigSMILES article.<sup>30</sup>

## 2.5. Nested Stochastic Objects for Multistep Polymers

G<sup>2</sup>RINS implements the description of nested stochastic objects that represent polymers created from prepolymers through multiple polymerization steps. To the best of our knowledge, this is the only line notation capable of describing this kind of structure while encoding

sufficient information for the complete representation of polymer materials and the generation of sampled polymer ensembles. This is a key feature because the vast majority of real-life polymer materials are synthesized and processed through complex multi-step pathways. To describe such polymers, G<sup>2</sup>RINS locates stochastic objects as repeat units or initiators within higher-order stochastic objects, as shown in the schematic G<sup>2</sup>RINS strings in Figure 2. Such objects are called nested stochastic objects. In practice, each stochastic object in this nested structure represents a step of polymerization. This process is recursive; thus, there is no limit to the number of nested steps in a polymer structure. Although nested stochastic objects are already available in BigSMILES they are not implemented in its generative version, G-BigSMILES. In addition, the grammar of nested stochastic objects is refined in G<sup>2</sup>RINS to account for the accurate representation of polymers created through real multi-step processes.

In G<sup>2</sup>RINS, nested stochastic objects can act as initiators or repeat units. Each option represents a different polymerization method. An initiator stochastic object is used to represent a process in which a prepolymerized structure serves as the starter in a subsequent chain-growth polymerization reaction; thus, the prepolymer chain occurs only once in the resulting polymer molecules. Figures 2b and 2c show examples illustrating the usage of a stochastic object as an initiator. Example 2b represents a graft polymer in which the backbone is synthesized in the first step, and then the side chains are attached in the second step, producing separate statistical distributions of the backbone and the resulting graft polymer. Example 2c represents a diblock copolymer in which block A is synthesized in the first step, producing a mixture of A chains with its corresponding molecular weight distribution, and block B is attached in a second polymerization step, after which a molecular weight distribution of the diblock is obtained.

A repeat unit stochastic object is used when a prepolymerization step is followed by a step-growth polymerization step or, more generally, when prepolymerized chains are used as repeat units in a subsequent polymerization reaction. Thus, prepolymerized chains may

appear several times in the resulting polymer structures. Figure 2d shows a polymer architecture represented by this type of stochastic object. In this example, the G<sup>2</sup>RINS string describes a multiblock polymer synthesized through the step-growth polymerization of pre-polymerized A and B chains. Both prepolymers A and B are represented in the string by stochastic objects with their own molecular weight distributions. These objects are located as repeat units of a higher order stochastic object that represents the multiblock ensemble with its molecular weight distribution. The resulting multiblock chains are random combinations of A and B chains sampled from their independent stochastic distributions.

Multiple nested stochastic objects acting as repeat units or initiators can be combined and recursively repeated to describe a wide variety of polymer architectures. More detailed explanations of how to represent polymers based on the basic structures shown in Figure 2 are available in Section 5 and Electronic Supporting Information (ESI).<sup>32</sup>

The implementation of nested stochastic objects is the most important achievement of G<sup>2</sup>RINS. The description of multi-step polymerization allows for the proper representation of the vast majority of polymer architectures that are currently not accessible to the existing line notation.

### 3. The G<sup>2</sup>RINS Graph

In this section, we will describe the graph representation associated with the G<sup>2</sup>RINS line notation. The G<sup>2</sup>RINS graph is a complete semantic translation of the string. The graph organizes the information stored in the string for convenient utilization in high-throughput computational endeavors. Our G<sup>2</sup>RINS implementation<sup>32</sup> automatically generates a graph from a G<sup>2</sup>RINS string. The package also handles the generation of chain ensembles by reading the graph and implementing a generative algorithm, as explained in Section 4.

A schematic drawing of the G<sup>2</sup>RINS graph is presented in Figure 3a. Each node in the graph represents an atom, as opposed to the G-BigSMILES graph<sup>30</sup> where each node

corresponds to a repeat unit. The node attributes store the atomic number and charge, as well as information important for the generation process, such as the ID of the stochastic objects to which the atom belongs and the corresponding molecular weight distributions. Atoms are connected by directional edges that represent different types of chemical bonds. The types are encoded as edge attributes, along with the connection probabilities. The static edges, colored gray in Figure 3, represent the fixed connections that are internal to a repeat unit, initiator, or terminator. These connections are not subject to the stochastic generation process; they occur automatically when the repeat unit is chosen to grow. The connections between repeat units inside a stochastic object are represented by propagation edges, colored red in Figure 3. These bonds have a certain probability of occurring during the stochastic growth of a chain according to the relative reactivity of the connection sites. They will occur several times in the resulting molecules. Transition edges, colored blue in Figure 3, connect the initiators to the corresponding repeat units. They can also represent the transition to a different stochastic object in a nested or non-nested structure. Finally, termination edges cap the growing chains. Transition and termination edges also have associated probabilities that reflect the site’s reactivity. In sum, static bonds encode fixed connectivity, transition bonds direct generative flow between units, propagation bonds continue chain growth, and termination bonds end an active growth path.

The graph is created to encode the information read from the source G<sup>2</sup>RINS string. The edges connect the repeat units, initiators, and terminators, following the connection rules and reactivities defined in the string. The nodes’ attributes encode the generative information present in the string, including the molecular weight distributions and the nested stochastic structure. Thus, the graph can be read as a generating object by following its directional edges from the source nodes (initiators) to the terminators.

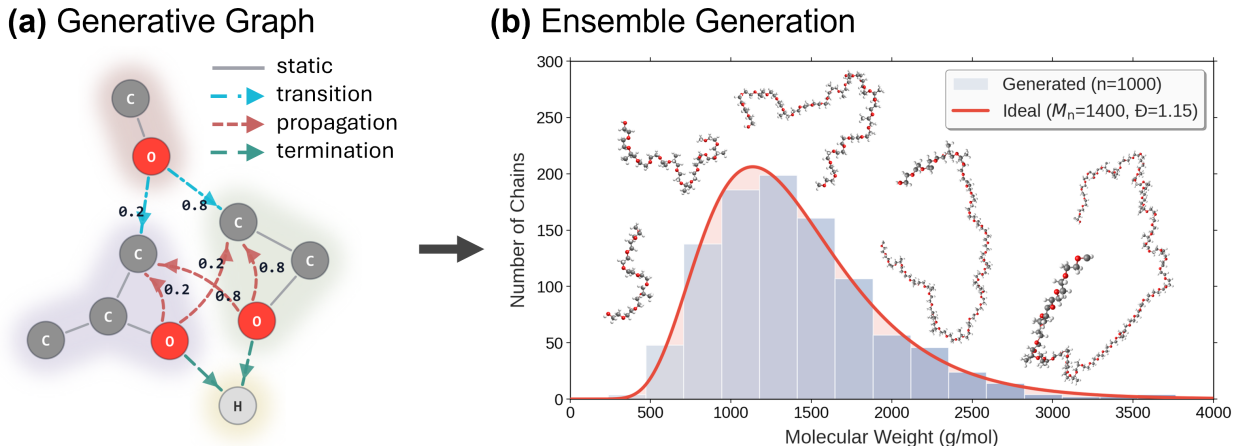


Figure 3: (a) Representation of the G<sup>2</sup>RINS graph for the example of Figure 1, represented by the G<sup>2</sup>RINS string

```
{[] [ <|.8|]CCO[>|.8|], [ <|.2|]CC(C)O[>|.2|]; CO[>]; [ <] [H]
[]}]log_normal(1400,1.15)|
```

The colored shadows at the background are added for visualization purposes and group nodes belonging to the same unit, including initiators (light red), terminators (light yellow) and repeat units (light purple and light green). (b) Histogram of Molecular Weights of the molecules generated from the graph (bars), and the target ideal distribution (red line). Representations of some of the created molecules are shown as inserts.

## 4. Ensemble Generation

Our generation algorithm is a Monte Carlo procedure that samples a chain following the generation rules encoded in the G<sup>2</sup>RINS graph. The generation process starts with randomly selecting a source node from the available initiators. In this choice, the reactivity of each initiation site defined in the corresponding bond connectors (or molar amounts) is taken into account. In the example shown in Figure 3, the only starter available is a methanol group; therefore, all the chains generated from this graph will be initiated by it. Next, a target molecular weight,  $M_i^0$ , is randomly drawn from the molecular weight distribution.  $M_i^0$  defines the extent to which the sampled polymer molecule will grow.

The core of the generation process is the stochastic addition of repeat units. The initiator will attach the first repeat unit by randomly choosing a transition bond (colored blue). In the example, the probability of growing a CCO unit is 0.8, while that of growing a CC(C)O unit is 0.2. Next, the chain will grow by attaching subsequent repeat units, following the

rules and probabilities defined by the propagation bonds (colored red). In this case, each repeat unit has a probability of 0.8 of growing a CCO unit and 0.2 of growing a CC(C)O unit. The chain will grow until the target molecular weight is reached. At this point, the chain will terminate by following the termination edges that connect to the terminators. In this simple linear homopolymer, the last repeat unit will terminate by growing a bromine with a probability equal to 1.0.

The algorithm considers the molecular weight of the terminators as part of the chain molecular weight. To achieve this, at each generation step, we compute the total molecular weight of the current chain,  $M_i$ , as the molecular weight of the non-terminated chain plus the average molecular weight due to termination, considering the likelihood of each possible terminator,  $M_t$ . When  $M_i + M_t$  exceeds the target molecular weight,  $M_i^0$ , the process is finished and random terminators are chosen from the terminator list. In this way, we ensure that the termination truly reflects the terminator bond probabilities defined in the string while also accounting for the terminators in the chain molecular weight.

The summarized process is as follows:

1. Random choice of the initiator based on their bond probabilities.
2. Sampling of a target molecular weight,  $M_i^0$ , from the molecular weight distribution.
3. Stochastic growth of the molecule is achieved by sequentially adding repeat units based on connectivity and bond probabilities until  $M_i^0$  (including terminators) is reached.
4. Addition of terminators.

In more complex structures comprising nested stochastic objects, a similar process is followed with slight variations. The main difference is that a stochastic object can terminate by transition to a different stochastic object. Every time a repeat unit belonging to a new stochastic object is reached, a new target molecular weight is drawn from the corresponding molecular weight distribution. When the process reaches a repeat unit that belongs to more

than one stochastic object in a nested structure, the target molecular weights from all of the nested distributions are drawn simultaneously. In section 5, we show examples of G<sup>2</sup>RINS strings and graphs representing such complex structures.

The G<sup>2</sup>RINS package directly creates a sampled ensemble comprising a number of chains defined by the user. This ensemble will follow the correct statistical distributions of molecular weight, monomer sequence, and monomer ratio for each polymerization step and for the polymer material as a whole. The resulting architecture of each chain will also statistically reflect the synthetic path represented in the G<sup>2</sup>RINS string. Figure 3 shows the histogram of molecular weights of an ensemble of 1000 chains for the EO-PO copolymer. The histogram obtained coincides with the expected molecular weight distribution defined in the string.

In addition, our implementation directly provides aggregated ensemble information, such as the total number of each repeat unit and each bond occurring in the ensemble. This information can then be used for various computational tasks, including setting up automatic pipelines to perform MD simulations, training ML models for property prediction, and generating polymer fingerprints. This topic will be briefly discussed in Section 6.

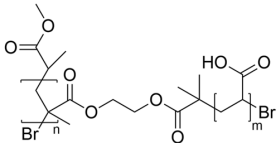
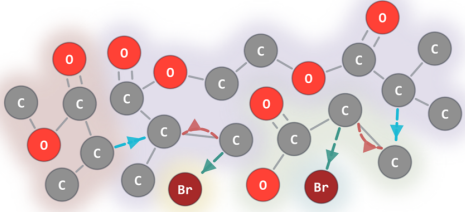
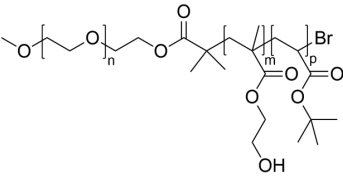
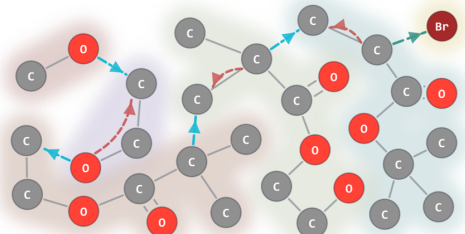
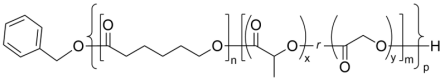
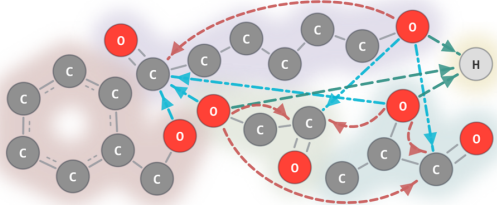
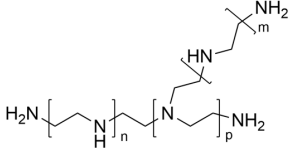
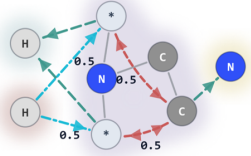
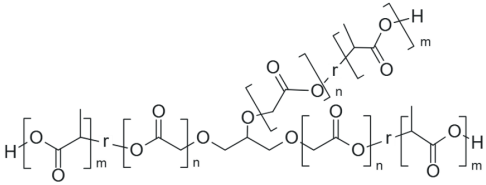
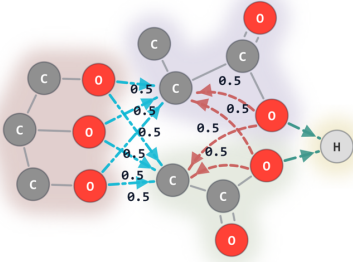
## 5. Representing Complex Polymer Architectures

In this section, we will show examples of complex architectures that G<sup>2</sup>RINS can represent using the new grammatical features developed in the notation. We will discuss graft polymers, nested block copolymers created by chain-growth and step-growth polymerization, branched polymers, and star polymers with multifunctional starters.

### 5.1 Graft Polymers

To represent graft polymers prepared by prepolymerization of the backbone and subsequent polymerization of the side chains ("grafting from" method),<sup>37</sup> G<sup>2</sup>RINS uses a nested structure in which the backbone stochastic object is defined as the initiator of the side chains.

Table 1: Chemical structure and G<sup>2</sup>RINS generative graph for five polymers illustrating the G<sup>2</sup>RINS description of graft, triblock, multiblock, branched, and star polymers. The colored shadows at the background are added for visualization purposes and group nodes belonging to the same unit, including initiators, terminators, repeat units, and linkers. Nodes belonging to the same unit are connected by static edges (gray). Linkers are units that occurs only once in the stochastic generation, usually attached to other units by transition edges. Repeat units comprise propagation edges (red), indicating the path for the unit repetition.

| Architecture      | Chemical Structure                                                                  | Generative Graph                                                                      |
|-------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <i>Grafted</i>    |    |     |
| <i>Triblock</i>   |   |    |
| <i>Multiblock</i> |  |   |
| <i>Branched</i>   |  |  |
| <i>Star</i>       |  |   |

— static — transition — propagation — termination

Table 1 shows the chemical structure of a graft polymer whose backbone is prepolymerized poly(2-hydroxyethyl methacrylate), PHEMA, and the side chains are poly(acrylic acid), PAA.<sup>38</sup> This example aligns with the schematic structure shown in Figure 2b.

The PHEMA backbone is represented by the following stochastic object:

```
{[] [<0]C(C)(C(=O)OCCOC(=O)C(C)(C) [>1])C [>0];  
COC(=O)C(C) [>0]; [<0]Br [<1]}|gauss(4000, 500)|
```

The repeat unit defined in the string has bond connectors with two different ID numbers. Bonds with ID 0, highlighted in red, represent the internal connectivity of the backbone. Bonds with ID 1, highlighted in blue, represent the pending connections to the side chains. The default ID number 0 is explicitly included in the string for clarity, although its inclusion is redundant. The structure of the end groups reflects the use of methyl 2-bromopropionate as the initiator of the atom transfer radical polymerization in the original synthesis.<sup>38</sup>

This stochastic object is the initiator of a higher order stochastic object that represents the whole graft polymer. The resulting G<sup>2</sup>RINS string is

```
{[] [<1]CC(C(=O)O) [>1];  
{[] [<0]C(C)(C(=O)OCCOC(=O)C(C)(C) [>1])C [>0];  
COC(=O)C(C) [>0]; [<0]Br [<1]}|gauss(4000, 500)| [>1];  
[<1]Br []}|gauss(8000, 1000)|
```

where the nested stochastic object representing the backbone is highlighted in red and the ID 1 bond connectors are marked in blue.

The corresponding graph is shown in Table 1 and reproduced in Figure 4a, where the different parts of the graft polymer are highlighted in different colors: the backbone, the graft junctions, and the side chain. The generation of a chain is schematized in Figure 4b. First, target molecular weights for both the PHEMA backbone and the molecule as a whole are drawn from the corresponding molecular weight distribution. Chain growth begins with

the PHEMA backbone, as depicted in step 2 of Figure 4b. The backbone generation will first follow its internal rules, including initiation, repeat unit propagation, and termination. After termination, the backbone end is capped with a Br atom, and the molecule has one open site per grown repeat unit. These open sites can initiate the growth of the PPA side chain with equal likelihood, as depicted in step 3 of the scheme in Figure 4b. This step is represented in the graph by a transition edge, colored blue in Table 1 and Figure 4a. Next, the PPA units grow from the backbone pending sites randomly, following the probabilities of connection defined in the string, as shown in step 4 of the scheme. After the first PAA unit is attached to a random pending site, the following units can either start a new side chain or lengthen the existing ones stochastically, as reflected in step 4 of the scheme. For this reason, it is possible that some of the grafting sites do not grow a side chain, reflecting the stochastic nature of the actual polymerization.

The final structure will have a molecular weight sampled from the outer distribution, `gauss(8000, 1000)`, while the backbone will have a molecular weight sampled from the inner distribution, `gauss(4000, 500)`. An ensemble of chains will asymptotically reproduce the molecular weight distributions of both the backbone and the entire polymer, as shown in Figure 4c. In addition to the molecular weight distribution, other important properties can be retrieved from the created polymer ensemble, such as the distribution of side chain degrees of polymerization (DP) or the grafting density per chain. For this polymer, the average side chain DP of the group is 2.39, while the average grafting density of the group per chain is 0.81. The grafting density is affected by the relative probability of connecting a side chain repeat unit to itself versus to the grafting point (red and blue arrows in step 4 of the scheme shown in Figure 4b), as well as the molecular weight distribution of the backbone and the overall polymer. A detailed analysis of the ensemble properties of this graft polymer is available in the ESI.<sup>32</sup>

G<sup>2</sup>RINS can also represent graft polymers prepared by the grafting-onto and grafting-through strategies<sup>39</sup> The grafting-onto strategy involves attaching prepolymerized side chains

to the backbone through a coupling reaction, while the grafting-through strategy consists of polymerizing side chains with reactive groups that are then connected to form the backbone in a second polymerization step. In the ESI,<sup>32</sup> we show examples of G<sup>2</sup>RINS representations of such graft polymers.

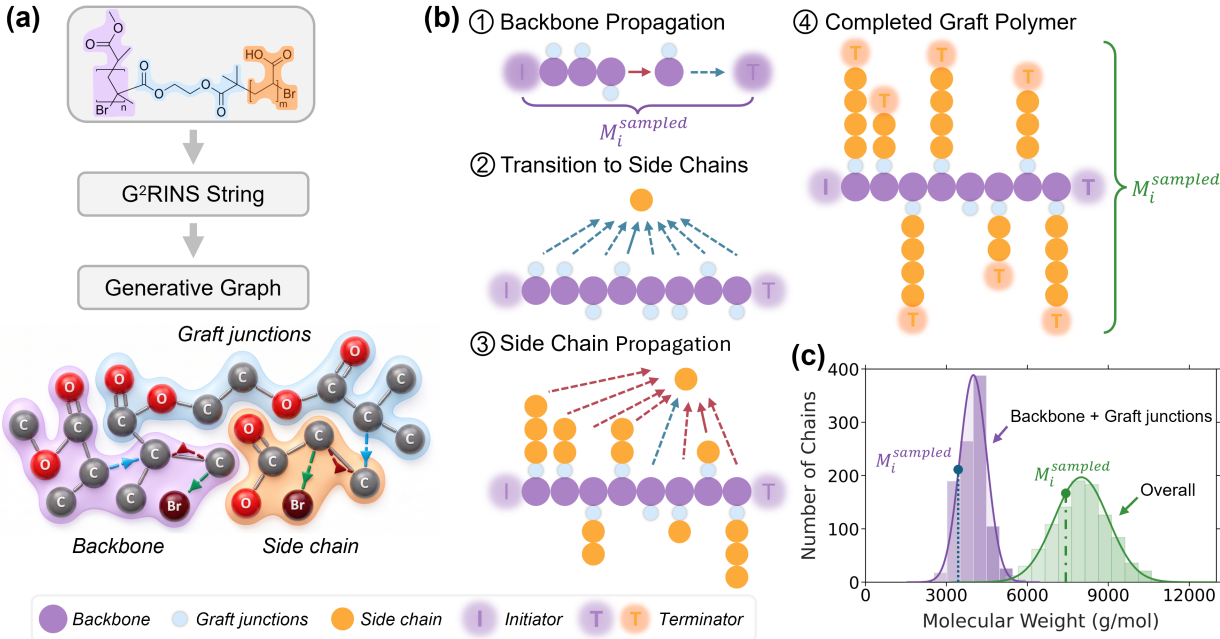


Figure 4: (a) Scheme of the conversion from the chemical structure to the G<sup>2</sup>RINS graph for the graft polymer from Table 1. The backbone, graft junction and side chain segments are highlighted in colors both in the chemical structure and the graph. (b) Schematic flowchart of the generation of one sampled molecule from the generative graph. (c) Histogram of the molecular weight of the backbone and graft junctions (purple bars), and the overall polymer (green bars) for a generated ensemble comprising 1000 chains. The solid lines are the target ideal distributions, which are Gaussian functions with number average molecular weight and standard deviation equal to 4000 g/mol and 500 g/mol for the backbone + graft junctions (purple line), and 8000.0 g/mol and 1000 g/mol for the overall polymer (green line).

## 5.2 Block Copolymers by Chain Growth Polymerization

The representation of block copolymers synthesized by multiple chain growth polymerization steps involves the recursive use of nested stochastic objects as initiators. Table 1 shows the chemical structure of an ABC triblock copolymer, poly(ethylene oxide)-block-poly(2-hydroxyethyl methacrylate)-block-poly(tert-butyl acrylate) (PEO-b-PHEMA-b-PtBA),<sup>40</sup> formed

by three consecutive polymerization steps. This example aligns with the basic structure depicted in Figure 2c. The corresponding G<sup>2</sup>RINS string is

```
{[] [<]CC(>)]C(=O)OC(C)(C)C;
  {} [<]CC(>)](C)C(=O)OCCO;
    {} [<]CCO>]; CO>]; [<]}|poisson(3000)|CCOC(=O)C(C)(C)>];
      [<]}|poisson(6000)|>];
    [<]Br []}|poisson(10000)|
```

where the PEO block is represented by the stochastic object colored red. This stochastic object acts as the initiator of the PHEMA block in a second-layer stochastic object colored blue (initiator in red) that represents the PEO-PHEMA diblock. Finally, this stochastic object is the initiator of the PtBA block in a third-layer stochastic object (black with the initiator in red and blue) that represents the entire triblock copolymer, PEO-PHEMA-PtBA.

The associated graph, whose representation is shown in Table 1, has transition edges connecting the blocks (colored blue). During generation, these transition bonds will be used only once and only after the corresponding growing block reaches its target molecular weight. A chain starts to grow from the CO unit. The PEO block will grow first. Since PEO is part of the three nested stochastic objects, all three target molecular weights of the three molecular weight distributions present in the string are drawn at this moment:  $M_{n(PEO)}$ ,  $M_{n(PEO-PHEMA)}$ , and  $M_{n(PEO-PHEMA-PtBA)}$ . The PEO block will grow until  $M_{n(PEO)}$  is reached. Next, a transition edge will be activated to connect the PEO block to the linker group  $C(=O)C(C)(C)$ , and then another transition edge will connect the linker to the PHEMA block. The PHEMA block will grow until the entire chain reaches the second target molecular weight  $M_{n(PEO-PHEMA)}$ . Finally, a similar procedure is followed to grow PtBA until the third target molecular weight  $M_{n(PEO-PHEMA-PtBA)}$ . When an ensemble of chains is created, all three molecular weight distributions are recovered, as shown in the ESI.<sup>32</sup>

Block copolymers created by a coupling reaction between polymerized blocks can also be represented by G<sup>2</sup>RINS by simply concatenating two stochastic objects. Such copolymers often have a linker between the two blocks that can be added as a SMILES group connecting both stochastic objects. ABA triblock copolymers created by the symmetric polymerization of external blocks, starting from an inner prepolymerized block, are represented through the use of multifunctional initiators. For instance, the string

```
{[] [<]CCO[>];
  {[] [<]CC(C)O[>]; O([>])[>]; [<]}|poisson(440)|[>];
  [<][H] []}|poisson(2200)|
```

represents Pluronic L44<sup>®</sup>, a poly(ethylene oxide)-poly(propylene oxide)-poly(ethylene oxide) (PEO-PPO-PEO) block copolymer.<sup>41</sup>

### 5.3 Multiblock Copolymers by Step Growth Polymerization

This example considers multiblock copolymers synthesized through a step-growth mechanism.<sup>42</sup> Table 1 shows the chemical structure of poly(L-lactide-*r*-glycolide)-*b*-Poly( $\epsilon$ -caprolactone) multiblock copolymers (PLGA-PCL MBCs).<sup>43</sup> This polymer is created in a two-step process: first, the PLGA-*b*-PCL chains are created with their own molecular weight distributions; then, PLGA-*b*-PCL chains are combined to form a multiblock copolymer (PLGA-*b*-PCL)<sub>*m*</sub> through a step-growth mechanism. This example corresponds to the schematic G<sup>2</sup>RINS grammar shown in Figure 2d.

The G<sup>2</sup>RINS string that represents this polymer is

```
{[] [<]{[>] [<]C(=O)CCCCO[>];; [<]}|gauss(2200, 1000)|[>1],
  [<1]{[>] [<]C(=O)C(C)O[>], [<]C(=O)CO[>];; [<]}|gauss(1300, 600)|[>];
  c1cccc1CO[>]; [<,<1][H] []}|log_normal(20000, 2.39)|
```

where nested stochastic objects are used to represent PCL (red) and PLGA (blue) blocks.

These nested objects are located as repeat units in the stochastic object that represents the resulting multiblock copolymer.

During chain generation, subsequent chains of PCL and PLGA will be created following their own molecular weight distributions (represented as `gauss(2200, 1000)` and `gauss(1300, 600)` in the string). This process will continue until it reaches the target molecular weight sampled from the overall molecular weight distribution, `log_normal(20000, 2.39)`. When an ensemble of multiblock chains is created, the histogram of generated molecular weights follows the defined distributions, both for the inner PCL and PLGA blocks and for the resulting multiblock. To illustrate this, the ESI<sup>32</sup> shows the generated histogram of molecular weights and the ideal statistical distributions for the three stochastic objects.

## 5.4 Branched Polymers

Table 1 shows the chemical structure of branched polyethylene imine.<sup>35</sup> To describe this polymer, G<sup>2</sup>RINS uses multiple bond connectors attached to a single N atom to represent the multiple reactive sites of the amino group, a novel feature introduced in G<sup>2</sup>RINS. The resulting string is

```
{[] [<]CCN(<[>])<[>]; [<] [H]; [>]N, [<] [H] []}|poisson(1900)|
```

where the multifunctional atom N and its bond connectors are marked in red.

Reflecting multifunctional atoms on a graph is challenging because the reactivity and connectivity of a site are encoded as node attributes and these properties are stochastic by nature. To achieve the representation of multifunctional atoms, the G<sup>2</sup>RINS graph adds phantom nodes to represent reactive sites, tagged with an asterisk in the graph shown in Table 1. These nodes lack chemical identity and are removed from the molecule after it is generated.

During the generation of polymer molecules with repeat branches, more than one reactive site may be available at each growth step. When this is the case, the next growing site is

randomly chosen, considering the site probabilities stored as node attributes in the graph. The molecular weight histogram of the generated ensemble reflecting the ideal distribution, and representations of selected sampled molecules are presented in the ESI.<sup>32</sup>

## 5.5 Star Polymers through Multifunctional Starters

Unlike G-BigSMILES, G<sup>2</sup>RINS is capable of describing polymers initiated by multifunctional chemical groups from which more than one individual chain can grow. To illustrate this characteristic, Table 1 shows a star polymer initiated by glycerol, `[>]OCC(O[>])CO[>]`, a multifunctional group with three reactive alcohol groups.<sup>44</sup> The arms are copolymerized lactic acid and glycolic acid. The G<sup>2</sup>RINS string representing this polymer is

```
{[] [<]C(C)C(=O)O[>], [<]CC(=O)O[>];
  [>]OCC(O[>])CO[>]; [<] [H] []}
|schulz_zimm(1800, 1200)|
```

where the multifunctional initiator is colored red.

Table 1 shows the G<sup>2</sup>RINS graph corresponding to the example. The stochastic growth of the arms starts from any of the reactive sites present in the glycerol initiator. This is reflected in the graph as transition edges (colored blue) coming from the three reactive nodes and pointing to the repeat units.

After the first repeat unit is attached, the following can lengthen the existing arm either by using a propagation bond (red) or initiating a new one by using the remaining transition bonds (blue). The choice is weighted by the reactivities of the open sites, stored as probabilities in the nodes that represent the connecting atoms. The selected sampled molecules and the molecular weight histogram for this example are shown in the ESI.<sup>32</sup>

This example represents a star polymer synthesized in one step. G<sup>2</sup>RINS can also represent star polymers obtained by a coupling reaction between prepolymerized arms and a multifunctional group.<sup>45</sup> This is achieved by directly connecting the multifunctional initiator to independent stochastic objects representing the arms.

The examples presented in this section do not represent a comprehensive list of polymer microstructures accessible to G<sup>2</sup>RINS, but they comprise most of the basic tools for representing complex polymer architectures. G<sup>2</sup>RINS also retains all G-BigSMILES representational capabilities. For example, the G-BigSMILES reference implementation offers details on how to handle polymer mixtures.<sup>31</sup> In addition, the G<sup>2</sup>RINS repository<sup>32</sup> offers a wider variety of examples, as well as detailed explanations of grammar and the generative process.

## 6. Discussion

In this section, we highlight the potential of G<sup>2</sup>RINS as a powerful tool for digital polymer discovery and the management of polymer data. The next subsections are dedicated to briefly discussing the different areas of polymer informatics to which G<sup>2</sup>RINS can contribute.

### 6.1. Detailed Description of Polymer Materials

The G<sup>2</sup>RINS string provides a detailed representation of polymer materials, containing key information such as the complex connectivity and architecture that arise from multistep polymerizations and the molecular weight distribution of each polymerization step. It significantly outperforms BigSMILES and G-BigSMILES in representational capabilities. Many polymers with similar connectivity and chemistry are represented by the same BigSMILES string, despite differing in composition, architecture, or molecular weight distribution, and hence having dissimilar properties. G-BigSMILES is a step in the right direction towards a detailed description of polymers, but lacks the grammatical features to represent the architectural diversity that arises from complex polymerization processes. G<sup>2</sup>RINS overcomes these issues and has the potential to become the standard label for polymer materials in large databases, replacing less detailed notations such as BigSMILES and P-SMILES.

## 6.2. Automatic Setup of High Throughput Molecular Dynamic Simulations

Line notations such as SMILES are often used as input for the setup of MD simulations for small molecules.<sup>10,11</sup> Simulation workflows require the creation of a simulation box containing the desired molecules, among other tasks. For polymer simulations, the automation of the box setup is hindered by the absence of a representative line notation, as well as by the complications associated with the stochastic nature of polymers. Polymer simulation boxes are often prepared manually to some extent, and the ensemble of molecules often misrepresents the actual material in terms of molecular weight, composition, and sequence distribution, jeopardizing the quality of the simulated properties.

G<sup>2</sup>RINS is a great candidate as a line notation for input automatic workflows for MD simulations. The G<sup>2</sup>RINS implementation automatically creates statistically accurate ensembles of polymer chains that can then be located in the simulation box. The chains will follow the proper statistical distributions of molecular weight, composition, sequence, and architecture. This would not only enhance the accuracy of the simulated properties, but would also allow for the automatic exploration of design parameters. Altogether, we envisage that the incorporation of G<sup>2</sup>RINS into high throughput workflows for MD simulations can boost the creation of large datasets of simulated polymer properties for their use in Machine Learning and Artificial Intelligence.

## 6.3. Fingerprints and Descriptors

Numerical representations of polymer materials that encode their chemical and structural features, such as fingerprints and descriptors, are a popular choice for machine learning models due to their compatibility with various ML architectures.<sup>46,47</sup> However, the development of generalized automatic methods for polymer fingerprinting remains an open field.<sup>6</sup> G<sup>2</sup>RINS offers an opportunity to fill this gap. Accurate averaged structural and chemical data can be

extracted from the representative polymer ensembles generated by the G<sup>2</sup>RINS graph. This data can potentially be incorporated into curated numerical representations. Another possible avenue for the development of numerical fingerprints or descriptors is the featurization of the G<sup>2</sup>RINS graph itself.

## 6.4. Similarity Search

One of the open challenges in polymer informatics is obtaining universal metrics to assess the pairwise similarity of polymer materials. In the past, graph representations,<sup>48</sup> ensemble information,<sup>23</sup> and fingerprints<sup>49</sup> have been used to develop similarity scores for both small molecules and polymers. G<sup>2</sup>RINS offers various potential pathways to derive universal metrics for polymer similarity. The G<sup>2</sup>RINS graph is a detailed representation of polymer materials; as such, it can be used to develop numerical methods to assess differences in graph topology and features. In addition, the ensemble information generated by the G<sup>2</sup>RINS implementation could also be the basis for similarity metrics. A possible way to pursue this path is to use ensemble-based fingerprints to create the score.<sup>23</sup>

## 6.5. Machine Learning

The design of a proper polymer representation is a sensitive step in the development of ML models for classification, property prediction, and inverse design.<sup>46</sup> Due to their stochastic nature, the representation of polymers poses a greater challenge than that of regular molecules. Major efforts have been conducted towards the encoding of polymer features for various ML architectures, including variational autoencoders,<sup>50</sup> traditional and graph neural networks,<sup>22</sup> tree-based models,<sup>51</sup> among others. G<sup>2</sup>RINS offers unique opportunities for the incorporation of both the chemical structure and the hierarchical architecture of polymers into machine-readable inputs for advanced learning.

The G<sup>2</sup>RINS graph provided by our code is designed to be directly connected to ML architectures based on graph neural networks. Both the format and the data structure were

selected to suit the needs of such ML models. The G<sup>2</sup>RINS object output is a NetworkX MultiDiGraph that can be used outside the context of the G<sup>2</sup>RINS implementation, ready to be transferred to any existing GNN library.<sup>52,53</sup> The node and edge features have a fixed shape for any graph, easing downstream processing of the graph through the Machine Learning algorithm. We envisage that GNN architectures can be specifically curated to learn from the polymer information encoded in the graph. In addition, the G<sup>2</sup>RINS graph can be reduced to simpler graph representations to train existing models.<sup>18,54</sup>

Another possible approach is to take advantage of the generative nature of G<sup>2</sup>RINS and develop polymer embeddings based on chain ensemble information. This information can be used to enrich existing ML polymer representations or to create de novo numerical embeddings, such as fingerprints and descriptors, to input into state of the art ML models. Finally, the G<sup>2</sup>RINS string itself can be the model input, an approach already explored for P-SMILES and BigSMILES.<sup>9</sup>

## 6.6. Future Challenges

We have shown that G<sup>2</sup>RINS has the potential to play a central role in the future of polymer informatics. Figure 5 shows a flow chart for computationally assisted polymer material discovery, where the envisaged role of G<sup>2</sup>RINS is indicated. To pursue this vision, G<sup>2</sup>RINS faces specific challenges that will be discussed in the following paragraphs.

The G<sup>2</sup>RINS notation and implementation present some limitations that must be overcome on the path towards a universal polymer representation. Some of them are inherent to the G<sup>2</sup>RINS approach and have already been discussed in the G-BigSMILES article.<sup>30</sup> Briefly, G<sup>2</sup>RINS represents a simplified idealization of real polymer ensembles, described through a limited number of parameters that model bond reactivities and ideal molecular weight distributions. This is a deliberate design choice that aims to retain compactness and simplicity. However, there is room for improvement towards greater flexibility. As a first step in this direction, we introduced in G<sup>2</sup>RINS the possibility of defining the molar amounts of

repeat units, initiators, and terminators, unlocking an optional extra control parameter for ensemble generation. In the future, we plan to incorporate non-idealized numerical molecular weight distributions obtained from experimental characterization as metadata.

In addition to the inherent idealization of chain ensembles, there are certain limitations specific to the current implementation that will be addressed in future versions. Some features of the BigSMILES notation have not yet been implemented in G<sup>2</sup>RINS, such as the description of ladder polymers and non-covalent interactions.<sup>55</sup> In addition, the inclusion of cross-links in the G<sup>2</sup>RINS notation, graph representation, and generation algorithm represents a major challenge due to the complex topological dependencies of such bonds. Finally, some ensemble behaviors arising from the chemical mechanisms underlying polymer synthesis, such as chain compositional drift,<sup>56</sup> are not captured by the current generation algorithm.

The G<sup>2</sup>RINS package takes a user-provided G<sup>2</sup>RINS string as input and outputs the corresponding G<sup>2</sup>RINS graph and generated chain ensembles, together with aggregated ensemble information. Dedicated methods associated with G<sup>2</sup>RINS must be developed, both upstream and downstream. These include, but are not limited to, automatic converters from established molecular formats such as molfile<sup>57</sup> or SMILES to G<sup>2</sup>RINS strings; fingerprinting methods and the derivation of similarity scores; the automated generation of simulation boxes and associated force-field parameterization based on G<sup>2</sup>RINS-generated chain ensembles; and the development of curated ML frameworks that take the G<sup>2</sup>RINS graph and other G<sup>2</sup>RINS-derived polymer embeddings as input. Some of these methods are being developed by our group and collaborators and will be available soon.

## 7. Summary

We presented G<sup>2</sup>RINS, a generative string and graph representation designed to capture the hierarchical and stochastic nature of polymer materials. G<sup>2</sup>RINS encodes sufficient in-

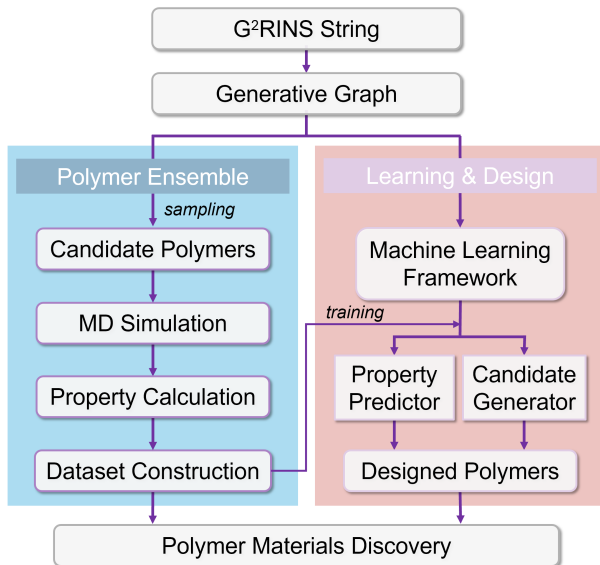


Figure 5: Flowchart of computational assisted polymer materials discovery, where the G<sup>2</sup>RINS role is highlighted.

formation to distinguish polymers with diverse chemistry, connectivity, molecular weight, and synthetic pathways. From the information stored in the string, G<sup>2</sup>RINS constructs a generative graph specifically designed for polymer informatics. Due to its generative nature, G<sup>2</sup>RINS produces ensembles of polymer chains that reflect the actual statistical distributions of molecular weight, composition, sequence, and architecture. In this way, G<sup>2</sup>RINS is effectively a multilevel polymer representation comprising the G<sup>2</sup>RINS string, the generative graph, and the generated chain ensemble.

G<sup>2</sup>RINS represents a significant improvement over existing representations. Traditional line notations such as SMILES and P-SMILES are limited to single, well-defined molecular structures or oversimplified polymer descriptions and therefore cannot capture the stochastic and architectural complexity of polymer ensembles. BigSMILES introduces stochasticity and connectivity, but primarily provides a qualitative description of polymer microstructure and lacks direct support for generating statistically consistent ensembles. G-BigSMILES extends this concept to enable generative capabilities, but remains limited in representing multi-step polymerizations and complex architectures.

G<sup>2</sup>RINS overcomes these limitations by introducing prepolymers as nested stochastic objects that explicitly encode multi-step synthetic pathways and hierarchical polymer architectures. This enables the rigorous representation of branched, grafted, star, and multiblock polymers, as well as their recursive combinations, within a single consistent formalism. More importantly, the nested representation is directly executable: each stochastic object can be generated and subsequently used as an initiator, backbone, or building block in a later polymerization step. As a result, G<sup>2</sup>RINS unlocks the accurate description of real-life polymers, capturing both their structural diversity and stochastic nature.

Beyond its representational power, G<sup>2</sup>RINS establishes a direct bridge between polymer chemistry and computational workflows. Thus, G<sup>2</sup>RINS has the potential to accelerate polymer materials discovery through applications in labeling, database queries, automated workflows for generating simulated polymer data, numerical representations, and machine learning models. Realizing this potential will require dedicated efforts to integrate G<sup>2</sup>RINS within the broader community and to develop methods that enable G<sup>2</sup>RINS-based high-throughput polymer informatics.

Overall, G<sup>2</sup>RINS represents a significant step forward towards a unified, machine-readable representation of polymers that captures both their chemical structure and their stochastic architecture. By enabling the consistent description and generation of realistic polymer ensembles, G<sup>2</sup>RINS provides a powerful foundation for advancing data-driven polymer design and accelerating computational material discovery.

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